CYBERSECURITY LOG ANALYSIS REPORT

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Activity: CTM-ACTIVITY-8

Date: NOVEMBER 8, 2026

Instructions: Analyze each CSV dataset provided in the GitHub repository. Identify the likely attack type, cite specific log evidence, assess whether the attack succeeded, and explain the possible impact if the attack were successful. Do not rely on filenames or guesses; support your conclusions with the log entries.

ATTACK 1 ANALYSIS

1. Identification of Attack

SSH brute-force and credential stuffing attacks both aim to gain unauthorized access to accounts by guessing login credentials

2. Evidence from the System Logs

Evidence 1: Event ID / Timestamp: BF-0001 / 2026-10-31T08:14:02Z
Observed activity: Failed SSH login attempt for username admin from source IP 198.51.100.23 on port 22
Why it matters: This represents the first attempt to gain access by trying commonly used administrative usernames and passwords.

Evidence 2: Event ID / Timestamp: BF-0004 / 2026-10-31T08:14:17Z
Observed activity: Multiple failed SSH login attempts were detected from 198.51.100.23 within 15 seconds, targeting high-privilege accounts such as admin, administrator, and root. This suggests a possible brute-force attack.
Why it matters: This shows repeated, rapid login attempts using different passwords to gain unauthorized access to privileged system accounts.

Evidence 3: Event ID / Timestamp: BF-0005 / 2026-10-31T08:14:22Z
Observed activity: The admin account was automatically locked after five unsuccessful login attempts, preventing further access attempts.
Why it matters: This shows that the system's security measures automatically stepped in to prevent further unauthorized login attempts.

3. Attack Analysis

An external attacker from IP address 198.51.100.23 attempted to gain unauthorized access to the internal server 192.0.2.10 through SSH (port 22). The attacker repeatedly tried different passwords for common administrator accounts, including admin, administrator, and root, making attempts every five seconds starting at 08:14:02Z. The system successfully detected and blocked the activity by locking the targeted account after five failed attempts (BF-0005) and applying rate-limiting by the sixth attempt (BF-0007), helping prevent further unauthorized access.

4. Attack Status

Mitigated / Blocked (Unsuccessful)

5. Possible Scenario if the Attack Succeeds

- The attacker successfully gained remote access to the server 192.0.2.10 and obtained administrative privileges, allowing them to interact directly with the system through a command-line shell.

6. Prevention and Mitigation

Recommendation 1: Disable password-based SSH login and require users to authenticate using SSH keys instead. Also, prevent direct root access by setting `PermitRootLogin no` in the SSH configuration to strengthen server security and reduce the risk of unauthorized access.

Recommendation 2: Set up automated protection using tools like firewall rules to detect repeated failed login attempts and automatically block suspicious IP addresses. This helps prevent brute-force attacks and reduces the risk of unauthorized access to the server.

Recommendation 3: Restrict SSH access to trusted connections through a firewall or VPN, allowing only authorized users to connect. Also, require multi-factor authentication (MFA) for all remote administrative logins to add an extra layer of security and prevent unauthorized access.

ATTACK 2 ANALYSIS

1. Identification of Attack

Credential Phishing Attack / Social Engineering

2. Evidence from the System Logs

Evidence 1: Event ID / Timestamp: PH-0002 / 2026-10-31T09:03:45Z
Observed activity: The security gateway detected a suspicious link (account-verify.example.test) in an email that appeared to come from it-support@example.test, raising concerns that the message could be a phishing attempt.
Why it matters: The link uses a different domain from the organization's official website, suggesting that the email sender may be impersonating the organization to deceive recipients and potentially steal sensitive information.

Evidence 2: Event ID / Timestamp: PH-0003 / 2026-10-31T09:05:18Z
Observed activity: The employee (employee01@example.test) clicked the suspicious link in the phishing email, potentially exposing their account or sensitive information to a security threat.
Why it matters: This shows that the phishing attempt successfully tricked the employee into clicking the malicious link by creating a sense of urgency through a fake password expiration warning.

Evidence 3: Event ID / Timestamp: PH-0004 / 2026-10-31T09:05:41Z
Observed activity: The browser or endpoint security system detected a fake login form on account-verify.example.test and blocked the page, preventing the employee from potentially entering their login credentials.
Why it matters: Confirms the destination URL was a fraudulent login interface designed to harvest corporate login credentials.

3. Attack Analysis

An attacker using IP address 203.0.113.80 sent a phishing email to employee01@example.test, pretending to be from the organization's IT support team. The email, titled "Urgent: Password Expiration," tried to pressure the employee into verifying their login credentials by falsely warning that their account would be disabled. The employee clicked the suspicious link at 09:05:18Z and was redirected to account-verify.example.test. Fortunately, the browser's security protection detected the fake login page and blocked access before any credentials could be stolen (PH-0004). The email gateway also successfully quarantined the phishing email (PH-0005), preventing further exposure.

4. Attack Status

Mitigated / Blocked (Unsuccessful)

5. Possible Scenario if the Attack Succeeds

The attacker successfully steals the employee's corporate username and password through the fake login page, potentially allowing unauthorized access to the organization's systems and accounts.

6. Prevention and Mitigation

Recommendation 1: Adds an extra layer of security, ensuring that stolen passwords alone are not enough for attackers to access company systems.

Recommendation 2: block suspicious emails that attempt to impersonate internal company addresses and reduce the risk of phishing attacks.

Recommendation 3: Provide regular security awareness training and simulated phishing exercises to help employees recognize suspicious emails, identify fake or unfamiliar domains, avoid clicking urgent or misleading links, and report potential phishing attempts to the IT or security team.

ATTACK 3 ANALYSIS

1. Identification of Attack

Distributed Denial of Service (DDoS) / HTTP Flood Attack

2. Evidence from the System Logs

Evidence 1: Event ID / Timestamp: DOS-0002 / 2026-10-31T10:20:10Z
Observed activity: Traffic from 198.51.100.90 suddenly increased to 420 requests per second, far above the normal rate of 35 requests per second. The system detected the unusual activity and triggered HTTP 429 rate limiting to restrict excessive requests and help prevent service disruption.
Why it matters: This indicates a sudden surge of excessive requests intended to overload the server and consume its resources, potentially disrupting normal service for legitimate users.

Evidence 2: Event ID / Timestamp: DOS-0004 / 2026-10-31T10:20:30Z
Observed activity: Traffic from multiple IP addresses suddenly increased to 850 requests per second, compared to the normal rate of 35 requests per second. This caused the service to slow down and return HTTP 503 (Service Unavailable) errors, indicating that the server was struggling to handle the unusually high volume of incoming requests.
Why it matters: Confirms multi-source distributed traffic overwhelming host capacity, leading to server degradation and service denial for legitimate users.

Evidence 3: Event ID / Timestamp: DOS-0005 / 2026-10-31T10:20:40Z
Observed activity: The automated system health check failed during the ongoing traffic surge because the service continued returning HTTP 503 errors, indicating that it was temporarily unavailable or unable to handle incoming requests.
Why it matters: This confirms that the web application became unavailable because the excessive traffic overwhelmed its resources, preventing it from responding properly to user requests.

3. Attack Analysis

Starting at 10:20:00Z, an attacker launched a coordinated HTTP flood attack against the web server at 192.0.2.20 through HTTPS port 443, causing traffic to surge from the normal rate of 35 requests per second to 850 requests per second from multiple IP addresses (198.51.100.90–198.51.100.93). The excessive traffic overwhelmed the server, resulting in severe slowdowns and HTTP 503 errors, which caused the automated health check to fail at 10:20:40Z. In response, the security system activated automated traffic filtering and rate limiting at 10:20:50Z (DOS-

0006), reducing incoming traffic to 160 requests per second. These measures successfully restored normal service operations by 10:22:00Z (DOS-0008).

4. Attack Status

Mitigated / Resolved

5. Possible Scenario if the Attack Succeeds

The web application becomes unavailable or stops responding, preventing legitimate users from accessing the service and potentially causing the organization to fail its promised service uptime commitments.

6. Prevention and Mitigation

Recommendation 1: Use a Web Application Firewall (WAF) and cloud-based DDoS protection services, such as Cloudflare or AWS Shield, to detect and filter excessive malicious traffic before it reaches the main servers. This helps protect the application from large-scale traffic floods, maintain service availability, and ensure legitimate users can continue accessing the system.

Recommendation 2: Configure edge proxies to automatically limit the number of requests allowed from each IP address or network within a specific period. Also, block or restrict suspicious traffic from untrusted IP ranges when unusual activity is detected. This helps prevent traffic overload, reduce malicious requests, and keep the application available to legitimate users.

Recommendation 3: Set up load balancers and automatic scaling to handle sudden increases in legitimate traffic by adding server resources when needed.

FINAL REFLECTION

1. Among the three attacks, which could cause the greatest damage to an organization? Explain your answer.

For me, Attack 2 could cause the greatest damage to an organization. Although we do our best to strengthen cybersecurity, we can never be completely certain that employees will not fall victim to phishing attacks, especially those who lack cybersecurity awareness. That is why one of my recommendations for Attack 2 is to conduct regular security awareness training and simulated phishing exercises. These activities will help employees recognize suspicious emails, identify fake or unfamiliar domains, avoid clicking urgent or misleading links, and report potential phishing attempts to the IT or security team.

2. Why is analyzing system logs important in detecting and responding to cybersecurity incidents?

Analyzing system logs is important because they help us keep track of what is happening inside a system. They show activities such as login attempts, access requests, errors, and unusual network traffic. By checking these logs, security teams can spot suspicious activities, identify possible cyberattacks, and understand how an attacker got into the system and what damage was caused. This allows organizations to respond to threats more quickly, prevent further attacks, and improve their security to avoid similar incidents in the future.

3. What is the most important cybersecurity lesson you learned from this activity?

These three attacks is that defense-in-depth and fast automation save the day when things inevitably go wrong. Looking at the incidents together, attackers targeted totally different weaknesses: abusing SSH logins with brute force, exploiting human trust through a fake password reset email, and trying to knock down a web application with a flood of junk traffic. What stands out most is that even when a real human mistake happened—like the employee clicking on that phishing link—it didn't result in a total disaster. That's because backup defenses like browser blocks, automated account lockouts, and smart traffic filtering kicked in right on time. At the end of the day, good security isn't about expecting perfection from your system or your team; it's about building layers of protection so that one slip-up doesn't tear down the whole house.

SUBMISSION CHECKLIST

☐ I analyzed all three CSV datasets.

☐ I identified each attack using evidence from the logs.

☐ I included at least three supporting log observations for each attack.

☐ I justified the status of each attack.

☐ I explained a possible successful-attack scenario for each attack.

☐ I provided at least three relevant security recommendations for each attack.

☐ I completed the final reflection.

Submission filename: Surname_Firstname_Cybersecurity_Log_Analysis.docx

Reminder: Distinguish between an attack attempt, detection, prevention, and actual compromise. A suspicious event or a clicked link does not automatically prove that an account or system was compromised. Base your conclusion on the available evidence.